

Whose Regulation Does the Model Know? Reliability of LLM-Retrieved Regulatory Information for Public Environmental Management in 25 Countries

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Abstract

Public administrations increasingly consult large language models about the regulations they are responsible for applying, yet no established method exists for auditing whether those answers match the binding text. We contribute such a protocol and report what it finds. It scores answers against official registers, adjudicating by code wherever the correct answer is a published value and by a three-vendor judge panel elsewhere, and we apply it to air pollution regulation across 25 countries, 14 deployment stacks, five task types and four languages (9,251 responses), evaluating each model *as served*. All three remedies an agency can try fail, and the most intuitive one makes accuracy worse: querying in the country’s own language *lowers* accuracy by 4.8 percentage points (Wilcoxon $p = 3 \times 10^{-15}$) and makes the model 17.7 points less likely to return a register-checkable value at all; a public-manager

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persona does not narrow the gap ($p = 0.27$); and a regionally tuned model ranks last of fourteen. Accuracy also collapses precisely on retrieving the binding value: 0.37, against 0.62 for synthesis. A North/South gap persists (+5.1 pp, permutation $p = 0.020$), concentrated where answers depend on a national fact — on the national standard, Global South odds are 0.22 of Global North odds. Scope is deliberately bounded: we measure agreement with published regulation in one domain, not general model capability, and identify the corpus channel only where the design has resolution. The protocol transfers to any regulated domain whose correct answer is published.

Keywords: algorithm auditing, geographic bias, multilingual bias, digital government, Global South

1. Introduction

Large language models (LLMs) have moved from research curiosity to working infrastructure of public administration in under five years: an OECD survey of core government functions documents generative-AI adoption across policymaking and public-service delivery, including deployments that help public-administration staff search regulations and draft technical summaries (OECD, 2024). Environmental agencies, municipal secretariats, and applied policy analysts increasingly draft technical syntheses, translate regulatory norms, locate official monitoring data, and characterize institutional frameworks with such assistance. Air pollution policy is a domain where this adoption carries direct stakes for public health. Fine particulate matter (PM_{2.5}) is the leading environmental risk factor for premature mortality worldwide (GBD 2019 Risk Factors Collaborators, 2020; World

[Health Organization, 2021](#)), and the burden falls disproportionately on the Global South ([GBD 2019 Risk Factors Collaborators, 2020](#)). When a public environmental manager in São Paulo, Lagos, or Jakarta asks an LLM about an ambient air quality standard, an attainment deadline, or the epidemiological evidence behind an intervention, an inaccurate answer is not a benign error. It can propagate into a regulatory brief, a procurement decision, or an emergency response during a pollution episode.

The question this paper addresses is whether LLMs serve Global South air pollution policy as reliably as they serve the Global North, and whether the way a query is framed changes the answer. Two recent strands of evidence make this question urgent rather than hypothetical.

Geographic factual bias is real and measurable.. [Moayeri et al. \(2024\)](#) documented $1.5\times$ higher factual error rates for Sub-Saharan African countries versus North American countries across 20 models and 11 World Bank indicators. [Manvi et al. \(2024\)](#) showed that LLM zero-shot ratings on subjective topics correlate strongly with income level (ρ up to 0.70 against socioeconomic conditions). [Mirza et al. \(2024\)](#) found that GPT-4 systematically privileges statements about the Global North over the Global South in factual verification tasks. These results establish the phenomenon, but none of them targets a single high-stakes applied domain with verifiable official ground truth, and none asks whether the bias can be modulated by prompt framing.

Linguistic and corpus asymmetry is structural.. [Joshi et al. \(2020\)](#) mapped a six-class hierarchy of linguistic resource availability in NLP, showing that the languages of most Global South countries sit in the under-served lower

classes. Under-representation in the training corpus is the dominant explanation for downstream geographic bias (Bender et al., 2021). This asymmetry is not incidental; it reproduces longstanding patterns of uneven digital and epistemic infrastructure that decolonial scholarship has analysed for AI systems (Mohamed et al., 2020).

We identify three gaps that current work leaves open for a domain like air pollution policy.

Gap 1: no benchmark targets air pollution policy with official ground truth.. Existing evaluations use numeric World Bank recall (WorldBench), subjective ratings (Manvi et al.), journalistic fact-checking (Global-Liar), or everyday cultural knowledge (BLEnD (Myung et al., 2024)). None measures the recall and synthesis tasks a public environmental manager actually delegates to an LLM: stating a binding ambient standard, retrieving an officially measured concentration, summarizing the health-burden evidence, identifying the legal instruments and executing agencies, or recommending an operational response. These tasks anchor to authoritative sources (national environmental councils, state monitoring agencies, the World Health Organization, the Global Burden of Disease estimates), which makes accuracy verifiable in a way that trivia benchmarks are not.

Gap 2: persona framing is unexamined as an experimental lever.. Practitioners routinely prefix queries with a role (“As a municipal environmental secretary, ...”), and they do so on the advice of the model vendors: Anthropic’s documentation tells developers to give the model a role in the system prompt (Anthropic, 2026) and OpenAI’s asks for an identity block stating the assistant’s purpose (OpenAI, 2026), while the practice is catalogued as

the Persona Pattern in the prompt-pattern literature ([White et al., 2023](#)). Whether adopting the persona of a public environmental manager *reduces* the Global South accuracy gap (by steering the model toward policy-relevant, regulation-compliant responses) or *amplifies* it (by inviting confident fabrication of regulatory specifics the model never learned) has not been tested in a pre-specified design. We treat persona as a within-subject experimental factor rather than an uncontrolled confound.

Gap 3: the corpus-representation mechanism is asserted, not tested for this domain.. The link between corpus representation and accuracy is assumed but rarely measured against confounders, and rarely separated into its two channels. We distinguish *language-corpus* size (how much text exists in a country’s language: mC4 token volume ([Xue et al., 2021](#)), OSCAR size ([Abadji et al., 2022](#)), Joshi class) from *country-coverage* breadth (how broadly the corpus is about a country: Wikipedia and Wikidata coverage), and test which predicts air pollution policy accuracy after adjusting for Human Development Index — a distinction that matters because the geographic gap is measured in English.

Objectives and scope

We state the study’s boundaries so readers can judge whether the evidence applies to their situation. This study has three objectives, and each is bounded.

First, to establish a protocol for auditing regulatory information. The object is agreement between a model’s answer and the value published in the official register, for tasks an environmental agency actually delegates. We

are not measuring model capability in general, reasoning quality, or usefulness for any purpose other than retrieving and summarising regulation in force. The protocol is the contribution that transfers; the domain is where we demonstrate it.

Second, to test the mitigations available to an administration. A public body facing poor answers about its own country can realistically do three things without building anything: declare the official role in the prompt, ask in the national language, or procure a regionally tuned model. Each has a procurement or workflow cost. We test all three as designed factors rather than discussing them. We do not test retrieval augmentation, fine-tuning, or human-in-the-loop verification, which require capacity that most agencies in our Global South sample do not have.

Third, to locate where reliability differs and, as far as the design allows, why. We report the distribution of accuracy across countries, tasks and languages, and we test the corpus-representation account at the level where the measurement has resolution. We do not claim causal identification of any mechanism, and we distinguish throughout what the design demonstrates from what it cannot.

Throughout, our unit of analysis is the model *as served* — a stack of model weights, vendor, and serving infrastructure reached through a pinned API or inference endpoint — rather than an idealised set of weights, because a public manager interacts with the served stack and any measured gap is a property of that stack (Section 3). Hypotheses, sampling and analysis plan were fixed before data collection (Nosek et al., 2018) for 15 countries and then extended to 25; the plan was never deposited publicly, so we do not

claim pre-registration, and the study is reported as exploratory throughout.

Contributions of this work

Our contributions are:

1. *An auditing protocol for regulatory information, not a trivia benchmark.*

The question an administration faces is not whether a model writes well about policy but whether the value it returns matches the one published in the official register. We build five task types from the everyday work of an environmental agency (Table 1), each keyed to an official source, so correctness is adjudicable rather than a matter of expert impression. The protocol transfers to any regulated domain where the correct answer is published: tax, licensing, public health, procurement.

2. *The three localisation fixes agencies reach for, tested rather than assumed — and all three fail.* Persona framing, native-language prompting and a regionally tuned model are each a procurement or workflow decision, and each circulates as received wisdom. We test all three as designed factors. None works, and the one a manager is most likely to try first does active harm: asking in the country’s own language lowers accuracy by 4.8 percentage points, most sharply for the language with the scarcest corpus.

3. *Model choice as an administrative decision.* Whether an agency runs an open-weight model on its own infrastructure or a vendor-hosted one is now an explicit procurement question (Robinson, 2026). We evaluate fourteen deployment stacks across that spectrum, treating the unit of

analysis as the model *as served*, since an agency contracts a served stack and not a set of weights.

4. *A sampling design that separates who is poorly served from why.* Countries are stratified along independent axes — UNCTAD North/South (operationalising coloniality (Mohamed et al., 2020)), the Joshi et al. linguistic-resource taxonomy (Joshi et al., 2020), and development indicators — so geographic, linguistic and economic explanations can be told apart instead of conflated. Each implies a different remedy and a different responsible actor.
5. *A mechanism test that reports its own limits.* Between countries, coverage and development are collinear over 25 observations and cannot be separated. They separate once the unit of variation changes: comparing tasks that depend on a national fact against tasks that do not, within the same country, identifies country coverage as the operative channel, and it survives among the nine countries whose official language is identical. We also flag that the pre-specified corpus proxy is defective in a way worth knowing for anyone who has used it — it assigns the English Wikipedia to every sample country with English as an official language, indexing colonial history rather than corpus size.
6. *Ground truth adjudicated by code, and what that changes.* Where a task’s correct answer is a number in an official register, the verdict is computed against that register rather than delegated to a language model, because comparing a value with an authorised range is exact and a judge can only reproduce it with noise. On the same responses, rebuilding two task keys from official registers and moving the compar-

ison into code cut the development gradient from $\rho = 0.51$ to $\rho = 0.37$ and more than doubled the native-language penalty. In this literature the ground-truth key, not the sample size, is the binding constraint.

7. *Open resources.* The full protocol, prompts, ground-truth registries with their official sources, analysis code, and the deviation log will be released under CC-BY-4.0 and MIT licenses.

2. Related Work

Prior work runs along four threads: the measurement of geographic factual bias, multilingual and cultural extensions, mitigation attempts, and LLM use in policy and health decision contexts.

2.1. Geographic factual bias

[Manvi et al. \(2024\)](#) established that frontier LLMs carry systematic biases against regions of lower socioeconomic conditions, with Spearman correlations up to 0.70 between model-generated ratings and a proxy for those conditions. The outcome space is subjective ratings, which leaves open whether the same bias appears in objective recall of regulatory facts. [Moayeri et al. \(2024\)](#) introduced WorldBench, quantifying factual recall against World Bank indicators across 20 LLMs and documenting 1.5 $\times$ higher absolute relative error for Sub-Saharan Africa than North America, and enabling automatic detection of citation hallucination. Its design is bounded by the macro-economic indicator space; an environmental manager asks narrower, source-specific questions — a national ambient standard, an attainment deadline, a city-year concentration — whose ground truth lives in environmental councils and monitoring agencies rather than one aggregated database. [Mirza](#)

[et al. \(2024\)](#) extended the question to temporal stability, showing that newer GPT-4 versions do not monotonically improve and that the privileging of Global North statements persists.

These studies establish that the phenomenon exists and is measurable. None isolates a single applied domain with verifiable official ground truth, and none treats prompt framing as a manipulable factor.

2.2. Multilingual and cultural extensions

[Myung et al. \(2024\)](#) covered 16 countries and 13 languages in everyday cultural knowledge, finding that LLMs perform better in the native languages of high-resource cultures and worse in those of low-resource ones — the finding that motivates our language hypothesis (H2) and connects to the resource hierarchy of [Joshi et al. \(2020\)](#). Regional benchmarks have proliferated: BRoverbs ([Almeida et al., 2025b](#)) for Brazilian proverbs, TiEBE ([Almeida et al., 2025c](#)) for regional historical events, and ALBA ([Vieira et al., 2026](#)) and AMALIA ([Simplicio et al., 2026](#)) for European Portuguese. These document concrete failures in culturally grounded tasks but do not address applied policy use. Work on Portuguese corpus construction ([Almeida et al., 2025a](#)) underscores how Common Crawl snapshots under-represent Lusophone content, the corpus asymmetry H4 tests.

2.3. Mitigation, persona, and role conditioning

[Opuszek and Böhm \(2026\)](#) found that reasoning helps in aggregate but does not eliminate regional disparities, and [Kerche et al. \(2026\)](#) proposed a place-based typology situating geographic bias in a broader sociotechnical landscape. Both are conceptual and aggregate.

Role conditioning is not a folk practice but what the vendors instruct their own users to do: Anthropic documents role setting in the system prompt (Anthropic, 2026), OpenAI’s guide opens the recommended message structure with an identity block (OpenAI, 2026), and the pattern is catalogued in the prompting literature (White et al., 2023). The systematic evidence on whether it helps is sobering: across four LLM families and 2,410 factual questions, expert personas did not improve accuracy over a no-persona control and slightly *reduced* it across 162 roles (Zheng et al., 2024), and an independent replication agrees (Basil et al., 2025). Whether persona conditioning helps or harms factual reliability in high-stakes domains is therefore unsettled and, to our knowledge, untested for geographic equity. We treat persona as a pre-specified experimental factor and test both the Norm-Compliance hypothesis (persona reduces the gap) and the Persona-Vacuum hypothesis (persona amplifies it).

2.4. LLMs in policy, health, and the administration

Le Coz et al. (2025) evaluated LLM policy recommendations for homelessness across four cities, finding “context-blind rigidity” in which models apply stable internal heuristics poorly calibrated to local realities and align more closely with Global North experts. That result covers one domain and one task type; we generalise to a regulatory and health domain with five task types spanning recall, synthesis and recommendation. In health, Kozlakidis et al. (2026) argue that hallucinations carry acute risk in regulatory environments with limited technical capacity, where language and cultural mismatch increases error — a risk that meets the mortality burden documented by the Global Burden of Disease estimates (GBD 2019 Risk Factors Collaborators,

2020).

The deployment question is no longer hypothetical. [Kim \(2026\)](#) estimates bureaucratic productivity effects of generative AI quasi-experimentally, and [Robinson \(2026\)](#) documents that choosing which model an agency runs is now an explicit procurement decision between open-weight and vendor-hosted options. Both our model-level findings speak to it: the openly available frontier models trail the closed ones, and the regionally tuned Portuguese model is the weakest of the fourteen, so the two choices an agency might make on sovereignty or locality grounds are the two that cost the most accuracy.

Two strands frame why a wrong answer matters administratively. [Choroszewicz \(2026\)](#) shows that generative-AI support tools position frontline practitioners as “moral crumple zones”, absorbing responsibility for failures originating upstream — exactly the position of a civil servant who files a brief citing a superseded emission limit. And [Cordella and Gualdi \(2024\)](#), analysing Italy’s intervention on ChatGPT, argue that technology-neutral regulatory frameworks do not reach the specific harms these systems produce, which leaves verification at the point of use as the operative control. The literature on reception is correspondingly developed: [Aoki et al. \(2024\)](#) test whether the type of explanation changes perceived accuracy, fairness and trustworthiness among those affected.

What none of this establishes is whether the underlying answer is *correct*, and whether correctness is distributed evenly across the states that rely on it. Perceived trustworthiness and measured accuracy are different quantities, and a system can be well explained, well governed, well received, and wrong. That is the gap this study addresses, in a domain where the correct answer

is published in an official register and can be checked.

2.5. Positioning of the present work

Our design differs from prior work in four ways: (i) a single high-stakes applied domain with verifiable official ground truth, rather than numeric trivia or cultural knowledge; (ii) persona as a pre-specified within-subject factor rather than an uncontrolled framing choice; (iii) theory-driven country sampling on independent axes (UNCTAD coloniality, Joshi linguistic resource, development) rather than regional convenience; and (iv) an explicit corpus-representation mechanism test with sensitivity analysis. No prior study combines a regulated public-health domain, a persona manipulation, and a pre-specified mechanism test.

3. Methods

3.1. Research design

We specify a pre-specified, factorial, multi-country benchmark experiment to quantify geographic and persona-modulated bias in 14 large language models on air pollution policy tasks relevant to public environmental management. The pre-specified design crosses 15 countries (stratified along three theoretically grounded axes), 14 LLMs spanning five access tiers, one theory-driven domain (air pollution policy), 5 task types, and 2 persona conditions (neutral versus public environmental manager), with 2 stochastic replications per prompt-model-persona cell; a post-hoc extension (Section 3.3) enlarges the sample to 25 countries. The stimulus set crosses each country’s five tasks and two personas in English, plus a within-country native-language

rendering for the nine countries with a target native language, scored on the $[0, 1]$ composite. The study protocol fixes, in advance of data collection, the hypotheses, primary models, persona protocol, exclusion criteria, and multiple-comparison corrections. A Brazil-only collection executed under an earlier three-domain design is reclassified as an extended pilot and is reported separately; it informed prompt calibration but contributes no confirmatory observations.

3.2. Country selection

Countries were selected via theoretical sampling (Patton, 2015) triangulated across three independent classifications. First, the UNCTAD Global North/South classification (UNCTAD, 2024) operationalizes the coloniality-of-knowledge framework that motivates this study (Mohamed et al., 2020). Second, the Joshi et al. (2020) six-class taxonomy of linguistic-resource representation in NLP corpora provided an orthogonal stratification dimension (Joshi et al., 2020). Third, World Bank income groups (FY26) (World Bank Group, 2025) captured economic position independently of the previous two axes.

The final sample of 15 countries — Brazil, Mexico, Argentina, Peru (Latin America); Nigeria, South Africa, Kenya, Egypt (Africa); India, Indonesia, Bangladesh, the Philippines (Asia); and the United States, Germany, Japan (Global North reference) — covers four world regions and four World Bank income groups. The linguistic-resource axis is far less differentiated than the sampling frame intended: the dominant official languages of these countries fall into only three Joshi classes, and twelve of the fifteen share Class 5. Extending to 25 countries does not help, since eighteen of the twenty-five

are Class 5. This is a property of the world, not of our draw, because the official languages of most states are themselves high-resource; but it means the linguistic axis has too little variance to be separated from the others, and we treat the Joshi results accordingly (Section 4.6). The twelve Global South countries also vary in domestic air quality governance maturity, which is relevant to ground-truth availability (Section 3.9). Countries with compromised statistical governance (e.g., active conflict) or without a publicly retrievable national ambient air quality standard were excluded; Oceania and Joshi Class 0 languages were not represented given feasibility constraints and are acknowledged as scope limitations.

3.3. Sample expansion after the pre-specified analysis

After the pre-specified 15-country analysis, and *transparently reported here as a post-hoc extension rather than part of the locked protocol*, the sample was enlarged to 25 countries to increase statistical power for the gradient (H1) and corpus-mechanism (H4) tests and to multiply the within-country language pairs available for H2. Ten countries were added under the identical prompt-construction, ground-truth, collection, and scoring procedures. Six are Global North references chosen purely to broaden the high-resource end of the gradient — the United Kingdom, Canada, Australia, South Korea, France, and Italy. Four are native-language-pair countries that add Spanish (Colombia, Chile) and Portuguese (Portugal, Angola) contrasts for H2; of these, Portugal is also Global North while Colombia, Chile, and Angola are Global South. The two roles overlap, so by tier the extension adds seven Global North countries (the six references plus Portugal), raising the Global North cell from three to ten, and three Global South, while by language

it raises Spanish-speaking pairs from three to five and Portuguese-speaking pairs from one to three. Because this extension was decided after observing the initial results, every effect computed on the 25-country sample is reported alongside its pre-specified 15-country counterpart, and no pre-specified conclusion is revised on the basis of the extension alone; the expansion is treated as a power-and-robustness check, not as a new confirmatory test. Angola, like Nigeria and Argentina, has *no* national ambient PM_{2.5} standard, providing additional no-standard cases in which a faithful model must decline to state a non-existent threshold rather than fabricate one.

To multiply H2 pairs without adding new languages, two additional question variants per (country, task) were also generated for the five original native-pair countries via a high-quality generator model and rendered in both English and the native language; these variants are used *only* for the paired English-versus-native contrast (H2), in which any imperfection in the generated ground truth cancels between the two members of a pair, and are excluded from absolute-accuracy comparisons (H1, H3).

3.4. *Country covariates*

For each country we compiled developmental and corpus-representation covariates. Human Development Index (HDI) (UNDP Human Development Report 2023–24, 2022 data) and GDP per capita index developmental position and enter the mechanism analysis as adjustment covariates.

The corpus-representation exposures are organized along a distinction that the mechanism analysis (H4) makes explicit, because the two channels apply to different effects. Language-corpus measures quantify how much pretraining text exists *in* a language and are the candidate mechanism for

the native-language penalty (H2): the mC4 token count per language (Xue et al., 2021, Table 6), the OSCAR-2301 deduplicated corpus size per language (Abadji et al., 2022), and the Joshi linguistic-resource class (Joshi et al., 2020). Country-corpus measures quantify how much the encyclopedic corpus is *about* a country, independent of language, and are the candidate mechanism for the geographic gap (H1/H4): the byte length of the English Wikipedia article on the country, the number of Wikipedia language editions with an article on it (Wikidata sitelink count), and the number of Wikidata statements on the country entity.

One point deserves stating plainly, because the two roles are easy to conflate. Wikipedia and Wikidata are never used as ground truth here. They are the *explanatory* variable of the mechanism test: measures of how much the training corpus is likely to contain about a country. What a model’s answer is scored against is always an official register — the national gazette for the binding standard, WHO databases for measured concentrations and health burden, and the UNEP legislative assessment for policy instruments (Section 3.9). Had we scored answers against an encyclopedia, the study would measure agreement with an encyclopedia, which is not the question a public administration faces.

The two corpus families are also not interchangeable: because the geographic gap is measured *in English* (language held constant across countries), a residual North/South gap cannot be a language-corpus effect, so H4 isolates the country-representation channel for H1 and the language-corpus channel for H2. All covariate values, official sources, and access dates are versioned in the analysis release.

3.5. Model selection

Fourteen LLMs were benchmarked across five access tiers, spanning roughly two orders of magnitude in parameter count and four training-data geographies (United States, China, Europe, Brazil). Tier A (open-weight frontier, 5 models) comprised large open-weight systems accessed via free inference endpoints. Tier B (open-weight mid, 3 models) and Tier C (open-weight small/regional, 2 models) were run locally via Ollama under pinned model tags; Tier C includes a Brazilian-Portuguese instruction-tuned open model (Cabra-Mistral 7B v3) used for the regional-model test (H3) against a scale-matched globally trained open model. Tier D (closed accessible, 2 models) and Tier E (closed frontier, 1 model) were accessed via official APIs. Each model was queried with an identical version string or pinned tag throughout the collection window; tags and, where available, weight hashes are recorded in the protocol release to guard against mid-collection model updates. The complete model roster with vendor, parameter count, access route, and execution backend is provided in Supplementary Table S1.

A note on the unit of analysis. We compare *models as served* through a fixed access stack (official vendor API for closed models; a pinned inference endpoint for open-weight models), not idealized model weights in isolation. Provider infrastructure, serving configuration, and API layer are therefore part of what each label denotes, and a residual gap could in principle reflect the stack rather than the weights. We hold this constant by pinning version strings/tags and the maximum-determinism configuration available per provider throughout a fixed collection window, and treat “model” throughout as shorthand for this served-model stack; disentangling weights from infras-

tructure is a separate question we do not claim to resolve.

3.6. Domain and task taxonomy

The single domain is air pollution policy, chosen because it anchors to officially published, verifiable ground truth (national ambient air quality standards, environmental-agency monitoring data, federal control programs) and maps to concrete decision activities of a public environmental manager. Five task types operationalize the domain:

The five tasks were not written as a generic knowledge quiz. Each corresponds to a question a public environmental manager has to answer in the course of ordinary work, and each was drafted so that a wrong answer would carry an administrative consequence rather than merely lose a point. Table 1 gives the tasks with the prompt exactly as it was issued to the models, using Brazil as the worked instance, alongside the decision each one stands in for.

The tasks are ordered by how much of the answer is fixed by an official source. T1, T2, and T4 admit verifiable ground truth and carry the primary accuracy signal; T3 and T5 are rubric-scored open-generation tasks, where competent answers legitimately differ. That ordering is what makes the task dimension informative rather than decorative: it lets us ask whether models fail uniformly or fail specifically where a published national value has to be recalled, which is the question the results turn on. The country-by-task mapping of official source classes, equivalence criteria, and verifiability status is specified in the ground-truth registry (Section 3.9), and the full 25-country prompt set is released with the data. The persona condition (Section 3.7) prefixes a manager frame to these same questions without altering a word of them.

Table 1: The five tasks, the prompt as issued, and the decision each stands in for. Prompts are shown for the Brazil instance under the neutral persona; every country receives the same five templates with its own entities substituted (city, agency, regulation).

Task	Prompt as issued (Brazil)	Decision it stands in for
T1 Technical standard	“What is the current national annual mean ambient air quality standard for PM _{2.5} in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences.”	Citing the binding limit in a licence condition, an enforcement notice or a technical opinion. The wrong value voids the act.
T2 Local factual datum	“According to the national or subnational environmental agency, what was the annual mean PM _{2.5} concentration in São Paulo, Brazil, in the most recent reported year? Provide value in $\mu\text{g}/\text{m}^3$, the reference station or network, and the source.”	Establishing the baseline a local air quality plan or a public report is written against.
T3 Health-evidence synthesis	“Summarize the epidemiological evidence on mortality attributable to ambient PM _{2.5} exposure in Brazil over the last decade. Include estimated annual deaths, primary causes, and at least one authoritative source.”	Justifying why air quality should be prioritised against competing demands on the same budget.
T4 Policy instruments	“List the principal national programs or legal instruments for monitoring and controlling ambient air pollution in Brazil, with the executing agency for each.”	Identifying which programme to draw on and which agency holds the mandate.
T5 Applied recommendation	“A metropolitan region of Brazil faces a forecast episode with PM _{2.5} expected to exceed $75 \mu\text{g}/\text{m}^3$ within 24 hours. Recommend three short-term policy actions (deployable within 48 hours) to reduce population exposure, each justified by public-health evidence or operational feasibility.”	Choosing the operational response to an acute episode, under time pressure.

3.7. *Persona manipulation*

Persona is a within-prompt factor with two levels, and the manipulation is not an arbitrary one: assigning a role to the model is the practice the vendors themselves document and recommend. Anthropic instructs developers to set a role, stating that even a single sentence changes behaviour (Anthropic, 2026); OpenAI’s guide asks for an “Identity” block describing the assistant’s purpose and role before the task itself (OpenAI, 2026); and the practice is catalogued as the Persona Pattern in the prompt-pattern literature (White et al., 2023). The neutral condition presents the query as a plain information request. The `public_manager_env` condition prepends a fixed role frame identifying the requester as a municipal or regional environmental management official, holding the substantive question identical across conditions. For Brazil’s T1 the frame is the single sentence *“I am the municipal environmental management official of São Paulo, Brazil, and I need to brief decision-makers.”*, placed before the unchanged question.

One implementation detail matters for how the result should be read. Every request in this study is sent as a single user turn with no system prompt, so the role frame is prefixed to the user message rather than installed as a system instruction, which is where the vendor guides place it. This is deliberate. The system-prompt route requires API access and a developer to configure it; the public servants whose decisions motivate this study reach these models through the consumer chat interface, where the only available channel is the message they type. We therefore test role framing as the sector actually performs it, and Section 5.7 states plainly what remains untested. A single role-frame template is applied uniformly across countries and tasks

(translated where the prompt language is non-English), so that any between-condition difference is attributable to the persona frame rather than to question wording. Persona order within each prompt-model cell is randomized, and a manipulation check verifies that the role frame is acknowledged in the response. The exact wording, translation procedure, and manipulation-check coding are pre-specified and reproduced in Supplementary Material. Persona is the basis of hypothesis H6, which tests whether the role frame reduces (or amplifies) the country-level accuracy gradient.

3.8. *Prompt construction and validation*

Prompts were authored for the air pollution policy domain by the research team with domain supervision (PPGSAU/UTFPR) and validated against the ground-truth registry before entering the confirmatory stimulus set. The analysis plan called for a 50-prompt pilot scored by two independent human raters, with Krippendorff’s $\alpha \geq 0.70$ required before scaling the design. *That pilot was not run*, and no rater-level α exists for this study. Rubric reliability is instead evidenced after the fact, on the confirmatory data, by a three-vendor judge panel over every item that required a judge (Section 4.11), which returns $\alpha = 0.527$, below the threshold the plan had set. We report this as a departure from the plan rather than as a threshold met. All prompts were authored in English and, for a stratified subset of countries, additionally in a relevant native language (sparse multilingual matrix; Supplementary Section S3); native-language renderings were produced by a high-quality LLM translator (Claude Sonnet 4.6) prompted to preserve the meaning, the persona frame, and proper nouns and acronyms, with the factual ground truth held in English (the target value is language-invariant). We did not employ

human native-speaker translators, which remains a limitation (Section 5). We did, however, audit the renderings after the fact by back-translating all 90 native prompts with a different model and having two further models compare each back-translation against the English original (Section 4.4). Following the Brazil pilot validation, every factual claim in a ground-truth entry was required to resolve to an official source URL or a peer-reviewed DOI before the corresponding prompt was admitted, enforcing the project Citation Verification Protocol at the ground-truth generation stage rather than only at the manuscript stage.

3.9. Ground truth and the ground-truth registry

Ground truth for the verifiable tasks (T1, T2, T4) was drawn from official sources of record: national air quality standards and the agencies that issue them (T1), environmental-agency monitoring reports (T2), and federal policy programs with their executing agencies and legal instruments (T4). For T3, the reference is a curated set of peer-reviewed epidemiological studies with resolvable DOIs rather than a single number. Because the validity of the geographic gradient (H1) and the persona effect (H6) depends on ground truth being equally available and equally verifiable across all countries, we constructed a per-country ground-truth registry that, for each (country, task) pair, records the class of equivalent official source, the resolvable URL, the gold value or rubric key, the reference date, the human validator, and a validation flag. The registry design, cross-country equivalence criteria, and the explicit separation of an AI knowledge gap from a ground-truth-availability gap are documented in `docs/ground_truth_registry_design.md` and summarized in Supplementary Material. Entries are versioned by access date,

source URL, and a SHA-256 hash of the retrieved document. Where ground truth could plausibly have appeared in training data, a sensitivity subset is constructed from documents dated after each model’s training cutoff. Entries whose official source is unavailable or non-equivalent for a country are flagged and excluded from the primary accuracy comparison for the affected (country, task) cell, so that missing ground truth is never scored as a model error.

3.10. Data collection

API and local calls were executed at temperature 0.3 with deterministic seeds where available, with 2 replications per prompt-model-persona combination to estimate stochastic variability within the budget. Collection occurred within a fixed window to minimize model-update confounds, with the pinned tag, version string, and acquisition timestamp recorded for every call; because served-model behaviour can drift as vendors update infrastructure, the reported accuracies are point-in-time estimates relative to this window, and the full per-provider acquisition dates are released with the dataset. All responses were stored alongside complete request metadata (model version string or tag, timestamp, token counts, finish reason, persona condition) in a versioned storage layer with SHA-256 checksums. Response quality gates excluded truncated responses (`finish_reason` $\neq$ `stop`) and responses in a language divergent from the request; both were retained and analyzed separately as secondary outcomes.

Per-cell coverage is not uniform across the model roster: free and rate-limited endpoints (for example GPT-OSS 120B via a shared inference provider) returned fewer admissible responses than the metered APIs, so the

scored N ranges from roughly 300 to 780 responses per model. We report each model’s realized N alongside its score (Table 2) rather than silently dropping under-covered cells, and a full model $\times$ country $\times$ task coverage matrix, with the quota and exclusion reasons for every shortfall, is provided in Supplementary Material so that no absence is hidden.

3.11. Measures

Scoring assigns each task the most reliable instrument it admits, which is not the same instrument everywhere.

Where the answer is a number in an official register, code decides. For T2 (a measured city-year concentration) and T3 (an attributable-mortality estimate), the returned value is extracted and compared with the authorised range from the register by a deterministic scorer, with plausibility guards that reject implausible extractions — a four-digit year read as a death count, or a national population read as a mortality figure. Comparing a value with a range is exact, and a language model asked to do the same arithmetic can only reproduce it with noise. Code resolves 62.5% of T2 cells and 83.6% of T3 cells; the rest are those where no value could be extracted with confidence.

Where judgement is required, a three-vendor panel decides by its mean. The residual of T2 and T3, together with all of T4, is scored by Gemini 2.5 Pro, Claude Sonnet 4.6 and DeepSeek-V3, chosen for vendor diversity so that intra-vendor correlation does not inflate apparent agreement. T1 and T5 retain the scores of the original single judge (GPT-5-mini), T1 because it always had official ground truth and T5 because it is a rubric judgement against no register. Reliability is reported as Krippendorff’s α , the single-judge ICC(2,1) and, as the operative quantity, the ICC(2,3)

of the panel mean, all computed on every item the panel scored rather than on a calibration subset. An intraclass correlation answers a practical question — had a different judge scored the same answer, how much would the score have moved? — and averaging several judges raises it even when each agrees only moderately with the others, which is why the panel mean is the operative quantity and a single judge is not. One panel member (DeepSeek-V3) is also an evaluated model, and we test for self-favouring directly. No human gold-standard layer was collected; ground truth is anchored instead to official primary sources through a documentary audit (Section 3.9).

The primary composite combines five pre-specified subcomponents (Supplementary Section S6), each normalised to $[0,1]$: factual accuracy against the registry entry (weight 0.30), contextual completeness against a pre-validated rubric (0.25), citation quality against the DOI- and official-source-verified registry (0.15), calibration (0.15), and hallucination absence, coded binarily (0.15). Every headline effect is recomputed under equal weights and a PCA-derived weighting, and a conclusion counts as robust only if its direction is stable across all three; all headline effects are (Section 4.12). The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test.

3.12. Analytical strategy

The three primary tests are pre-specified as a single family (F1) and corrected together with the Bonferroni-Holm procedure, so a result surviving it is unlikely to be an artefact of testing several hypotheses at once. Secondary tests (F2) are reported unadjusted within family, and declared

exploratory tests (F3) use Benjamini-Hochberg FDR at $q = 0.10$ (Benjamini and Hochberg, 1995).

H1 (geographic gradient, primary). Country-level mean composite accuracy is regressed on the Joshi/HDI gradient via Spearman ρ , pre-specified one-sided with threshold $\rho \geq 0.55$ at $\alpha = 0.05$, with a Mann-Kendall trend test as a mandatory non-monotonicity complement and a Global South dummy as a sensitivity contrast.

H4 (corpus-representation mechanism, primary). Country-level partial Spearman ρ relates corpus representation to accuracy after adjusting for HDI and log GDP per capita, with Wikipedia counts as the pre-specified proxy ($\rho \geq 0.55$), complemented by exploratory country-coverage proxies (Wikidata sitelinks and statements).

H6 (persona effect, primary). The persona effect is a country-level difference-in-differences on the composite, $\Delta_{\text{GS}} - \Delta_{\text{GN}}$, where each Δ is the persona-minus-neutral mean within tier. H6 is supported (one-sided, $\alpha = 0.05$) if the persona condition reduces the Global South gap by at least 0.05 on the $[0, 1]$ composite. Inference uses a country-level permutation test (5,000 permutations) alongside a parametric paired test; the two-sided test is reported as a secondary check because persona amplification is theoretically possible.

H3 (regional model, secondary). A pre-specified one-sided Mann-Whitney contrast (Cliff’s δ) compares Cabra-Mistral 7B v3 against the other models.

H2 and H5 (exploratory). The prompt-language $\times$ country interaction (H2) and the open- versus closed-frontier contrast (H5) are reported descriptively, with formal tests under F3 correction where applied.

Three cross-cutting analyses corroborate the tier gap: a Gaussian linear mixed model (`statsmodels` MixedLM, REML) regressing the composite on the Global South indicator with crossed random intercepts for country and model, adjusting for centred HDI; a Bayesian hierarchical re-estimation of the same effect (`pymc` (Abril-Pla et al., 2023), weakly informative priors); and E-values for robustness to unmeasured confounding (VanderWeele and Ding, 2017). The corpus mechanism is probed with an exploratory country-level mediation model (`semopy` (Igolkina and Meshcheryakov, 2020)), and the persona manipulation with a role-frame-acknowledgement check on the response text. Pre-specified robustness analyses — leave-one-country-out, range restriction, and composite weighting — are reported in Section 4.12.

3.13. *Study protocol and open science*

The full protocol — hypotheses, prompts with hashes, the ground-truth registry, the persona template, scoring rubrics, mixed-model and DiD specifications, multiple-comparison corrections, and exclusion criteria — is pre-specified in advance of confirmatory data collection. The analytic dataset, after quality gates but before model fitting, is frozen and hashed so that the confirmatory analysis can be audited against the locked specification. The full protocol, prompts, ground-truth references, analysis code, and anonymized response data will be released on publication under CC-BY-4.0 (data and prompts) and the MIT License (code) (Rover et al., 2026); a companion Data Paper is prepared for *Scientific Data*.

3.14. Ethical considerations

No personal data from end-users of LLMs were collected. The study analyzes model outputs to publicly relevant policy questions; domain experts who contribute to ground-truth validation are acknowledged as co-authors when they meet ICMJE criteria. LLM provider terms of service were reviewed and respected, and responses are redistributed solely for research under fair-use and reproducibility provisions, excluded from any commercial application.

4. Results

Sample note. We report a pre-specified *15-country* benchmark and its *post-hoc extension to 25 countries* (the latter adds six Global North references — the United Kingdom, Canada, Australia, South Korea, France, Italy — and four native-pair countries — Colombia, Chile, Portugal, Angola). Fourteen LLMs are scored across five tasks, two personas, and four languages (English plus a country-native language — Spanish, Portuguese, or Hindi — for the nine applicable countries, giving the within-country English/native pairs that test H2). Every effect computed on the 25-country sample is reported alongside its pre-specified 15-country value; no pre-specified conclusion is revised on the extension alone. Scoring is adjudicated by code wherever the answer is a number checkable against an official register, and by the mean of a three-vendor judge panel elsewhere (Section 4.11); the T1 ground truth is anchored to official primary sources for all 25 countries (Section 4.9).

4.1. Sample and scoring

The analysed corpus comprises 9,251 scored responses on the $[0, 1]$ composite, of which 7,580 are English-prompt responses across the 25 countries

(the remainder are the within-country native-language pairs used for H2).

Scoring is assigned by the most reliable instrument each task admits. For T2 and T3, where the correct answer is a number published in an official register, the verdict is computed by code against that register: comparing a returned value with an authorised range is exact, and a language model asked to do the same arithmetic can only reproduce it with noise. Code resolves 62.5% of T2 and 83.6% of T3; the residual, together with all of T4, is scored by the mean of a three-vendor panel (Gemini 2.5 Pro, Claude Sonnet 4.6, DeepSeek-V3). T1 and T5 retain their original scores, T1 because it always had official ground truth and T5 because it is a rubric judgement with no register to check against. In total 5,373 cells carry a corrected score.

The panel mean replaces the single-judge design of the original protocol, in which the judge (GPT-5-mini) was itself among the evaluated models. Two measurements justify the change. Judges differ sharply in severity — mean composite 0.617 for Gemini, 0.451 for Claude, 0.379 for DeepSeek — so a single judge fixes an arbitrary level; and single-judge reliability is only $ICC(2, 1) = 0.56$ against $ICC(2, 3) = 0.79$ for the panel mean (Section 4.11). One panel member (DeepSeek-V3) is also an evaluated model, so we checked directly for self-favouring: DeepSeek scores its own outputs *more* harshly relative to the other two judges (-0.177) than it scores any other model’s (-0.123 to -0.160), so its own ranking is not an artefact of judging itself.

4.2. Accuracy by model

Table 2 orders the 14 models by mean composite. Frontier systems lead and small open-weight models trail. The result bearing on H3 is unambiguous: the Brazilian-Portuguese regional model Cabra-Mistral 7B is the

weakest of all models (0.347 versus 0.549 for the rest; Mann-Whitney $p \approx 0$, Cliff’s $\delta = -0.47$). Cliff’s δ has a direct reading: it is the probability that a randomly drawn answer from one group beats one from the other, rescaled to run from -1 to $+1$. A value of -0.47 means the regional model loses roughly three such comparisons for every one it wins. This contradicts the optimistic H3 framing that a regional model buys domain accuracy. Scale and frontier training dominate regional instruction tuning at the 7B level.

Table 2: Composite accuracy by model across 25 countries (code-adjudicated where the answer is a register value, panel mean elsewhere; $[0, 1]$). N = scored English-prompt responses.

Model	N	Mean
DeepSeek-V3	500	0.680
GPT-5	780	0.672
Gemini 2.5 Flash	490	0.624
GPT-5-mini	777	0.617
Claude Haiku 4.5	490	0.580
Qwen3 32B	498	0.553
Command R+	500	0.526
Llama 3.3 70B	474	0.507
Llama 4 Scout	500	0.492
Qwen3 14B	500	0.480
GPT-OSS 120B	298	0.479
Phi-4 14B	500	0.473
Llama 3.1 8B	777	0.437
Cabra-Mistral 7B	496	0.347

4.3. Accuracy by task: the recall floor

Difficulty is sharply task-dependent (Table 3). The hardest task by a wide margin is T1 (technical-standard recall, 0.367) — the task that requires returning the national annual PM_{2.5} standard in force. The open recommendation task (T5) scores highest (0.773) because its rubric rewards plausible, well-structured policy advice that does not hinge on a single fact. Pooling the two recall tasks (T1, T2; mean 0.417) against synthesis and recommendation (T3–T5, 0.623), the difference is large (Mann-Whitney $p \approx 0$, Cliff’s $\delta = -0.47$, again roughly three losses for every win). LLMs are weakest exactly where applied environmental management needs them most: recalling the precise regulatory threshold in force.

The correction of the ground truth is itself informative here, and provides an internal check on it. T1 always had official ground truth, and its score does not move at all (0.367 before and after). T2 was previously scored against a placeholder, and it rises from 0.368 to 0.467 once each returned concentration is compared with the authorised range by code. The correction therefore moved exactly the task whose ground truth was defective and left untouched the task whose ground truth was sound, which is the behaviour expected if the correction is right. It also revises the earlier reading: the floor is real but shallower than reported, and it is concentrated in normative recall (T1) rather than shared equally with local-datum recall (T2).

4.4. H2 — asking in the local language lowers accuracy

For the nine countries with a target native language (Brazil/Portugal/Angola → Portuguese, Mexico/Argentina/Peru/Colombia/Chile → Spanish,

Table 3: Composite accuracy by task type (25 countries). T2 and T3 are code-adjudicated where the returned value is checkable against the official register; T4 is panel-scored; T1 and T5 retain their original scores.

Task	N	Mean
T5 (applied recommendation)	1,508	0.773
T3 (health-evidence synth.)	1,519	0.560
T4 (policy instruments)	1,494	0.534
T2 (local factual datum)	1,530	0.467
T1 (technical standard)	1,529	0.367

India $\rightarrow$ Hindi), each prompt is rendered in both English and the native language with content held fixed, so the contrast is within-country and within-model. Replicates are averaged within each (model, prompt) cell before pairing, since pairing replicate-to-replicate would inflate n and make the test anticonservative. Across the 839 matched cells, native-language prompting *lowers* mean composite accuracy by 4.8 percentage points (Wilcoxon signed-rank $p = 3 \times 10^{-15}$).

The penalty tracks the resource level of the language, and all three are significant (Table 4): Hindi -11.1 pp ($p = 5 \times 10^{-4}$), Spanish -4.4 pp ($p < 10^{-4}$), Portuguese -3.9 pp ($p = 2 \times 10^{-4}$). India, which scores highest of all 25 countries in English, drops the furthest in Hindi.

Because H2 carries the study’s central claim, we tried to make it disappear rather than to confirm it, and it held under every attack. It is not driven by any one country (leave-one-out over the nine native-pair countries: -4.2 to -5.2 pp, all significant), by any one model (leave-one-out over fourteen: -4.2 to -5.1 pp, all significant), or by outlying cells (5% trimmed: -4.3 pp,

$p = 5 \times 10^{-19}$), and it survives every weighting of the composite (-4.7 pp under equal weights, -5.4 pp on the factual subcomponent alone). It appears in every task and under both persona conditions.

Two attacks deserve their own statement because they answer the objections a sceptical reader raises first. The first is that language-model judges might penalise non-English prose for style rather than content. We therefore test a binary outcome no judge touches: does the response contain a value the extractor can pull out and compare with the register? English prompts yield a verifiable value 66.4% of the time and native-language prompts 48.7%, a paired difference of -17.7 pp, with the native version worse in 122 of the 182 discordant pairs (sign test $p = 5 \times 10^{-6}$) and the same ordering by language. Asking in the local language does not merely lower a judged score; it makes the model substantially less likely to return a checkable number at all.

The second is that the native prompts, produced by an LLM translator, might simply ask something different. We tested this on all 90 unique native prompts — a census, not a sample — back-translating each with a *different* model and having two further models audit the result against the English original, with one flag enough to mark a prompt divergent. On the two items that could confound the effect, fidelity is complete: no prompt changed the country and none changed the requested quantity (90/90 on both). Restricting H2 to the prompts verified as faithful leaves it essentially unchanged (-4.5 pp, $p = 9 \times 10^{-8}$, $n = 541$) against -5.2 pp among the divergent ones, a difference not distinguishable from zero (permutation $p = 0.42$). Translation fidelity does not predict the size of the penalty, which is what one expects if translation is not causing it. Full audit output is in

Supplementary Material.

The effect is weakest in the code-adjudicated subset (-2.2 pp, not significant), and the reason is structural rather than contradictory: code adjudicates only where a number exists, so that subset conditions on the model having produced a value in *both* languages, which selects away the severest failure mode. It is also underpowered for an effect that size (power 0.18 at 2 pp, $n = 183$).

The applied implication is the opposite of the intuition. A manager who suspects an English-language model serves their country poorly, and responds by asking in the national language, makes the answer *worse* — and worse by the largest margin precisely where the language is least represented in training corpora. Taken with H6 below and the regional-model result above, all three localisation remedies fail together, and this one actively harms.

Table 4: H2: native-minus-English accuracy by native language (paired within country and model, replicates averaged within cell, 25-country sample).

Native language (Joshi class)	Countries	Δ (pp)	p	N cells
Spanish (5)	MEX, ARG, PER, COL, CHL	-4.4	$< 10^{-4}$	457
Portuguese (4)	BRA, PRT, AGO	-3.9	2×10^{-4}	311
Hindi (4)	IND	-11.1	5×10^{-4}	71
Overall		-4.8	3×10^{-15}	839

4.5. H6 — *persona does not narrow the geographic gap*

The persona factor was fully crossed, so H6 is tested as a difference-in-differences. The public-environmental-manager frame leaves the North/South gap essentially unchanged: $+4.7$ pp under the neutral frame

versus +5.4 pp under the manager frame, a DiD of +0.6 pp, far below the pre-specified 5 pp threshold; a country-level permutation test (5,000 permutations of the tier label) gives $p = 0.27$. H6 is not supported: presenting the query as coming from a public environmental manager neither narrows nor amplifies the Global South disadvantage, consistent with evidence that expert personas do not reliably improve factual accuracy (Zheng et al., 2024; Basil et al., 2025). A manipulation check finds that 50.2% of persona-condition responses explicitly acknowledge the role frame, so the null is not an artefact of the frame being universally ignored.

4.6. H1 — a tier gap that holds, a gradient that does not

H1 was pre-specified in two parts, and they now separate cleanly: one holds and one does not.

The tier gap holds. The ten Global North countries average 0.570 against 0.520 for the fifteen Global South countries, a gap of +5.1 percentage points whose 95% bootstrap confidence interval over countries is [+1.6, +8.5] pp and *excludes zero*. It is stable to dropping any single country (leave-one-out range [+4.5, +6.0] pp) and to the composite weighting (+4.2 pp under equal weights, +8.8 pp on the factual subcomponent alone, where it widens).

Where the gap lives, and why the composite understates it. The composite averages five tasks, and the gap is not spread evenly across them. Re-estimating the same contrast as a binomial mixed model on each task separately — outcome coded as returning the register value, with crossed random intercepts for country and model — shows the disadvantage scaling with how much the task depends on a fact specific to that country (Table 5). On the binding national standard (T1) the Global South odds of a correct

answer are 0.22 of the Global North’s (52.2% versus 29.4% raw), and on policy instruments (T4) 0.33; on health-evidence synthesis (T3), where the answer is an international estimate rather than a national one, the ratio rises to 0.69; and on open recommendation (T5), the one task with no register value to get right, the gap disappears entirely.

The gap is not weak — it is *concentrated*, and the composite dilutes it by averaging the tasks where it is severe with a task where it does not exist. On the question a public manager most often brings to these systems, what is the standard in force here, a Global South country is served at roughly a quarter of the odds of a Global North one.

Table 5: Global South versus Global North odds of returning the register value, by task. Binomial mixed model, crossed random intercepts for country and model.

Task	Answer is	GN	GS	OR	95% interval
T1 (technical standard)	a national value	52.2%	29.4%	0.22	[0.18, 0.27]
T4 (policy instruments)	national instruments	62.0%	44.7%	0.33	[0.26, 0.40]
T2 (local measured datum)	a national value	50.7%	37.8%	0.47	[0.39, 0.55]
T3 (health-evidence synth.)	an international estimate	59.1%	52.3%	0.69	[0.59, 0.80]
T5 (applied recommendation)	no register value	99.6%	99.7%	3.58	[0.84, 15.17]

The monotonic gradient does not. At $n = 25$, Spearman $\rho(\text{accuracy}, \text{HDI}) = +0.37$ does not reach significance ($p = 0.073$), and neither does the Mann-Kendall trend ($p = 0.072$). At the pre-specified $n = 15$ it is essentially absent ($\rho = +0.09$). No leave-one-country-out re-estimate reaches the pre-specified $\rho \geq 0.55$ threshold (range $[+0.31, +0.49]$), and no alternative weighting does either ($\rho = +0.31$ under both equal weights and the factual subcomponent). We therefore report a tier difference we can demonstrate and a development

gradient we cannot: the disadvantage is established as a difference between groups, while its shape across the development range remains exploratory. Under Bonferroni-Holm across the primary family {H1-gradient, H4, H6}, no test rejects its null.

The country ordering also retains large axis-orthogonal heterogeneity (Table 6): India (0.652, Global South) tops all countries, South Africa and Indonesia outrank most of the Global North, and Canada and France sit low. The disadvantage is a tier difference layered over substantial country-specific variation, not a clean ranking. This heterogeneity is itself part of why the gradient does not resolve.

Table 6: Composite accuracy by country, 25-country sample (tier: GN = Global North, GS = Global South). Ordered by mean.

Country	Tier	Mean	Country	Tier	Mean
India	GS	0.652	France	GN	0.536
Japan	GN	0.611	Bangladesh	GS	0.534
Australia	GN	0.604	Colombia	GS	0.530
United States	GN	0.592	Chile	GS	0.513
South Korea	GN	0.590	Canada	GN	0.510
Indonesia	GS	0.584	Mexico	GS	0.507
South Africa	GS	0.579	Kenya	GS	0.506
Italy	GN	0.577	Angola	GS	0.485
United Kingdom	GN	0.573	Nigeria	GS	0.461
Portugal	GN	0.563	Brazil	GS	0.459
Philippines	GS	0.546	Egypt	GS	0.455
Germany	GN	0.544	Argentina	GS	0.445
Peru	GS	0.538			

4.7. *H4 — corpus representation: country coverage, not language size*

The pre-specified mechanism for the geographic gap was training-corpus representation, proxied by Wikipedia article and edition count. That proxy is null: Spearman $\rho = +0.14$ ($p = 0.50$) uncontrolled, and still null partialling out HDI and log GDP per capita, the adjustment set the plan specified. The two control sets give the same answer because they carry nearly the same information (HDI and log GDP per capita correlate at $\rho = 0.97$ here). H4 as pre-specified is not supported. A post-hoc country-coverage proxy — the number of Wikipedia language editions with an article on the country — reaches $\rho = +0.36$ ($p = 0.075$) and attenuates after adjusting for HDI (partial $\rho = +0.18$, $p = 0.41$). At country level neither channel is established, because with 25 countries coverage and development are collinear.

Changing the unit of variation resolves what the country level cannot

The obstacle is confounding, not absence of signal, and it has a way out that does not require more countries. Each country answers five tasks that differ in how much the answer depends on a fact about *that* country: T1 (the national standard), T2 (a locally measured value) and T4 (national instruments) do; T3 (an international WHO estimate) and T5 (a generic recommendation) do not. Countries score 0.457 on the first block and 0.663 on the second, a specificity deficit of 0.206.

If what limits the model is how much the corpus says about a given country, that deficit should be smaller where coverage is greater — and a country’s HDI cannot explain a deficit measured *within* that same country. Estimating the interaction between country coverage and task dependence at the response level, with a fixed effect for country that absorbs development,

income, official language and everything else distinguishing countries, gives $\beta = +0.026$ (SE 0.005, $p = 2 \times 10^{-8}$, $n = 7,580$): greater coverage predicts a smaller deficit, in the direction the mechanism requires.

Four checks separate mechanism from artefact (Table 7). The *purest* contrast — T1, wholly national, against T5, wholly generic — yields the *largest* coefficient ($\beta = +0.042$, $p = 3 \times 10^{-12}$), the dose-response a real effect produces and a spurious association does not. Dropping T5, whose ceiling is 99.7%, or T4, the most interpretive, leaves it intact. A placebo inverting the labels returns the exact mirror image ($\beta = -0.026$). And entering both candidate channels together, country coverage remains strong ($\beta = +0.025$, $p = 4 \times 10^{-8}$) while the language-corpus proxy is marginal ($\beta = +0.010$, $p = 0.041$). That prediction was asymmetric and therefore falsifiable: because every response here was elicited *in English*, the size of a country’s local-language corpus has no obvious route to how much the model knows about that country, so had the language channel survived and coverage failed, our reading would have been wrong.

The resulting claim is narrower than “coverage causes the gap”. The design removes country-level confounding by construction, not confounding at the level of task-by-country. What we can say is that the country-coverage channel makes a differential prediction, that it holds, that it strengthens as the contrast sharpens, and that it survives adjustment for the rival channel — a mechanism supported at the level where this design has resolution, and what the country-level correlation was too confounded to show.

The corpus account also shows itself in H2, where the unit is the language rather than the country. There the direction is as predicted, though with

Table 7: Country coverage $\times$ task dependence, response-level mixed model with a fixed effect for country. Positive β = greater coverage, smaller deficit on country-dependent tasks.

Task classification	β	p
T1, T2, T4 versus T3, T5 (primary)	+0.026	2×10^{-8}
T1 versus T5 (purest contrast)	+0.042	3×10^{-12}
Excluding T5 (ceiling at 99.7%)	+0.021	4×10^{-4}
Excluding T4 (most interpretive)	+0.031	8×10^{-10}
Placebo: labels inverted	-0.026	2×10^{-8}

only three native languages it remains descriptive: the per-language penalty falls monotonically with mC4 token volume (Xue et al., 2021) and OSCAR size (Abadji et al., 2022) (Spearman = -1.00 over the three), with Hindi, the smallest corpus (mC4 24B tokens), carrying the largest penalty.

4.8. H5 — open versus closed-accessible

Grouping by access type, closed-accessible models average +13.3 pp above open-weight models. This runs against the optimistic H5 framing; the open arm includes larger models (Llama 3.3 70B, DeepSeek-V3), so the gap is not attributable to scale alone, although DeepSeek-V3 itself now ranks first overall. H5 is interpreted descriptively.

4.9. Ground-truth provenance

Because measured accuracy depends on the ground truth, the technical-standard target (T1) for all 25 countries was anchored to official primary sources by documentary verification rather than expert assertion, and is reproducible from the cited URLs — Brazil’s annual PM_{2.5} standard of

17 $\mu\text{g}/\text{m}^3$, for instance, read verbatim from Annex I of CONAMA Resolution 506/2024 in the Diário Oficial da União. Three countries have *no* national $\text{PM}_{2.5}$ standard, confirmed from the official text: Nigeria’s NES-REA regulation sets PM_{10} only, Argentina has no binding federal value, and Angola has no national air-quality legislation. The audit established this against circulating but incorrect secondary figures, and these are the cases in which a faithful model must decline to state a non-existent threshold rather than fabricate one.

The same standard covers the remaining tasks: T2 is adjudicated against WHO Ambient Air Quality Database v6.1 (23 of 25 countries have a city in the base), T3 against WHO GHO indicator AIR_41 with its uncertainty intervals, and T4 against Appendix 1 of the UNEP Global Assessment of Air Pollution Legislation, covering 24 of 25. Any reader can re-verify each value from the official source. The Wikipedia and Wikidata measures used in the mechanism test (Section 4.7) play the opposite role — they are the explanatory variable, and no answer is ever scored against them.

4.10. *Qualitative failure modes*

To make the composite numbers concrete, Box 1 shows three representative failures drawn verbatim from the response logs (run IDs available in the release).

The first mode is the most consequential for the no-standard cases (Angola, Nigeria, Argentina): rather than abstaining, models invent a plausible regulatory cutoff and disagree across replicates and models on the invented value — exactly the behaviour a public manager cannot detect without the official source.

Table 8: Box 1 — Representative failure modes (verbatim, lightly truncated).

Failure mode	Example
Fabricating a standard that does not exist	Angola has no national norm. Gemini 2.5 Flash asserted an annual standard of “25 $\mu\text{g}/\text{m}^3$ ” in one replicate and “15 $\mu\text{g}/\text{m}^3$ ” in another. GPT-5 asserted “15 $\mu\text{g}/\text{m}^3$ ” for Nigeria, which regulates PM_{10} only (NESREA). A faithful response would decline to state a value that does not exist.
Missing the binding value (the T1 floor)	Brazil’s official annual standard is 17 $\mu\text{g}/\text{m}^3$ (CONAMA 506/2024). The modal answer was 15 $\mu\text{g}/\text{m}^3$ (42 responses), with 10, 20 and 25 also common; the official value was almost never returned.
Degradation in the native language (H2)	India’s annual standard is 40 $\mu\text{g}/\text{m}^3$. GPT-OSS 120B returned the correct value in English and an <i>empty</i> answer to the same question in Hindi.

4.11. Inter-judge reliability

Reliability is reported on the sample that produces the effects, not on a calibration subset. All 3,190 items that required a judge were scored by all three panel members (Gemini 2.5 Pro, Claude Sonnet 4.6, DeepSeek-V3), giving Krippendorff’s $\alpha = 0.527$ (interval), single-judge $\text{ICC}(2, 1) = 0.558$, and — the operative quantity, since the analysis uses the panel mean — $\text{ICC}(2, 3) = 0.791$. Pairwise Pearson agreement is +0.80 between Claude and DeepSeek and +0.65–0.66 between each and the systematically more lenient Gemini. A single judge would carry $\text{ICC}(2, 1) = 0.56$, moderate at best, which is why the protocol specifies three vendors and analyses their mean.

The panel also isolates a design rule worth stating generally. Where the judging prompt asked the model to compare a returned concentration against a tolerance band expressed in words — that is, to perform the arithmetic itself — inter-judge concordance on those items was $r = -0.04$. Pre-computing the admissible range and asking only whether the value falls inside it raised concordance on the same items to $r = +1.00$, and the 274 scores collected under the ambiguous wording were discarded and recollected. Apparent judge disagreement can be an artefact of what the judge is asked to compute rather than a genuine divergence of judgement. Arithmetic belongs to code; the judge should be asked only what requires judgement.

4.12. Robustness

The two supported effects were tested against seven attacks. Both survive all of them; the development gradient fails all of them in the same direction, which is what makes its exploratory status a finding rather than an omission. Full outputs are in Supplementary Material.

No single country, and no weighting, carries the result. Leave-one-country-out re-estimation holds the tier gap in $[+4.5, +6.0]$ pp and the native-language penalty at -4.7 to -5.0 pp. Reweighting the composite does not move the conclusions either: under equal weights the gap is $+4.2$ pp and the language penalty -4.7 pp; restricting to the single most objective subcomponent — factual accuracy against the register value, discarding completeness, calibration and style — widens them to $+8.8$ and -5.4 pp, and deepens the recall floor (T1–T2 factual accuracy 0.370 against 0.657 for T3–T5). The effects are strongest where the measurement is most objective.

The tier gap survives a distribution-free test and two model families. A

permutation test states itself in a sentence: shuffle the North/South labels among the 25 countries ten thousand times over and count how often chance alone produces a gap this large. It assumes nothing about the distribution and treats the country, not the answer, as the independent unit. Chance clears the observed gap 2.0% of the time ($p = 0.020$). A Bayesian hierarchical model with crossed random intercepts agrees and puts the posterior clear of zero ($\beta = -0.052$, 94% HDI $[-0.092, -0.016]$, $P(\beta < 0) = 0.994$, $\hat{R} = 1.001$). A frequentist mixed model returns the same magnitude and sign but leaves it marginal ($\beta = -0.066$, $p = 0.069$), because it charges the large between-country heterogeneity to the error term. Two of the three tests clear zero and all three agree on magnitude, so we report the gap as supported and specification-sensitive, with an E-value of 1.70 at the point estimate and 1.31 at the confidence limit nearest the null ([VanderWeele and Ding, 2017](#)) — a country-level confounder of modest strength would suffice to explain it away, and the reader is entitled to weigh that.

The gap tracks world-system position, not national wealth. Global North and South follow the UNCTAD classification, and a gap visible under one taxonomy alone would be an artefact of it. Re-estimated under three development-based partitions of the same 25 countries (Table 9), the gap is +5.1 pp and significant under UNCTAD and smaller and non-significant under every split drawn on HDI. This coheres with the rest of the paper rather than undercutting it: the HDI gradient is itself null, so a split on HDI should not be expected to produce a gap. That UNCTAD separates the data where HDI does not points to position in the world system rather than income as the operative variable, which is what the coloniality-of-knowledge

framing predicts (Quijano, 2000; Mohamed et al., 2020). The reading is post hoc, and we therefore claim the gap for the North/South distinction as conventionally drawn, not as a general statement about development.

Table 9: Tier gap under alternative partitions of the same 25 countries. Permutation test with country as the unit, 10,000 relabellings.

Partition	Gap (pp)	GN/GS	p
UNCTAD North/South (used throughout)	+5.06	10/15	0.020
HDI below the sample median (0.781)	+2.39	13/12	0.269
HDI < 0.800 (UNDP “very high” cutoff)	+2.92	12/13	0.182
HDI < 0.700	+1.53	20/5	0.575

The null gradient is informative, within stated bounds. A null earns interpretation only to the extent the design could have seen the effect. Simulating 25-country samples at known correlations, this design has 77% power at $\rho = 0.55$ — the threshold fixed in advance — and a minimum detectable effect of $\rho = 0.56$ at 80% power. A development gradient of the magnitude we committed to calling substantial is therefore excluded; a weak one is not (power is 48% at $\rho = 0.40$). The ten countries added post hoc fall *within* the HDI range the original fifteen already spanned ($[0.548, 0.950]$), so the extension raises the density of observations rather than the range of the predictor.

4.13. Summary

Table 10 maps each hypothesis to its finding, the candidate mechanism, and an explicit evidence status, so no claim is read as stronger than the data support.

Table 10: Hypotheses, findings, candidate mechanisms, and evidence status.

Hypothesis	Finding	Mechanism	Evidence status
H2 language	-4.8 pp; Hindi -11.1	language corpus	Supported
H3 regional model	$\delta = -0.47$, worst of 14	scale over tuning	Supported
H1 tier gap	$+5.1$ pp; OR 0.22 on T1	tier	Supported
H1 gradient	$\rho = 0.37$, $p = 0.073$	development	Not significant
H4 within country	$\beta = +0.026$, $p = 2e-8$	country coverage	Supported, not causal
H4 pre-specified	language-corpus proxy null	language corpus	Not supported
H5 open/closed	$+13.3$ pp for closed	access type	Descriptive
H6 persona	DiD $+0.6$ pp, $p = 0.27$	—	Not supported

Note. H2’s mechanism rests on three native languages and is descriptive. H1’s tier gap is concentrated in tasks whose answer depends on a national fact. H4 within country removes country-level confounding by construction, not causal identification.

Three effects are supported with effect sizes and survive every robustness attack in Section 4.12: the native-language penalty, the recall floor on binding values, and the tier gap — the last modest on the composite but unambiguous where it matters, since on returning the binding national standard Global South odds are 0.22 of Global North odds ($[0.18, 0.27]$) and the gap vanishes on the one task with no register value to get right. Two pre-specified expectations are not met: the development gradient does not reach significance, and the persona frame does not narrow the gap. The corpus mechanism is identified within countries, where the design has resolution, and not between them, where coverage and development are collinear.

5. Discussion

We set out to measure whether LLMs answer air pollution policy questions as reliably for the Global South as for the Global North, whether prompt framing changes the answer, and what mechanism drives any gap. The benchmark — 25 countries, 14 models, four languages, five tasks, two personas, scored against official ground truth by code where the answer is a register value and by a three-vendor panel elsewhere — returns a disciplined and partly counter-intuitive picture. The harms we can demonstrate are practical rather than aetiological: the three remedies an agency would reach for all fail, the most intuitive of them makes the answer materially worse, and models are least reliable on the recall of the binding value in force. A North/South tier difference persists, and we identify the channel behind it only where the design has the resolution to do so.

5.1. Three intuitive remedies, all failing — and one that harms

The applied contribution is largely negative, and it is the most actionable result.

The local language does not help and materially hurts. Asking in the country’s own language lowers accuracy by 4.8 pp (Wilcoxon $p = 3 \times 10^{-15}$, $n = 839$ matched cells), significantly in all three languages, and the penalty scales with the scarcity of the language’s corpus: Hindi −11.1 pp, Spanish −4.4, Portuguese −3.9. India, the highest-scoring country of all 25 in English, falls furthest in Hindi. This is the mechanism the resource hierarchy predicts (Joshi et al., 2020; Xue et al., 2021), and the measurement has the resolution to see it because the unit is the language rather than the country.

The effect does not depend on any one country, model, task, persona condition or scoring instrument, nor on translation quality — a back-translation census of all 90 native prompts found none that altered the country or the requested quantity — and it holds on an outcome involving no judge at all: native-language prompts return a register-checkable value only 48.7% of the time against 66.4% in English (-17.7 pp paired, sign test $p = 5 \times 10^{-6}$). The harm is not that a judge scores non-English prose more harshly; the model becomes markedly less likely to produce a checkable number at all.

The local-manager persona does not narrow the gap (DiD $+0.6$ pp, permutation $p = 0.27$), consistent with evidence that expert personas do not reliably improve factual accuracy (Zheng et al., 2024; Basil et al., 2025). This null carries more weight than a null about a folk habit would, because role framing is the first thing every major vendor tells its users to do: Anthropic instructs developers to give the model a role in the system prompt (Anthropic, 2026), OpenAI’s guide opens the recommended message structure with an identity block (OpenAI, 2026), Google’s Gemini documentation asks that role definitions — which it calls persona — be placed in the system instruction (Google, 2026), and xAI’s standard example initialises every conversation with a role-assigning system prompt (xAI, 2026). Four independent companies converge on the same advice, and an administration following its supplier’s documentation writes the sentence we tested.

What that documentation claims for the sentence, read closely, is control over tone, format and framing — and nothing in it marks the boundary, so nothing tells a reader that the technique which reliably shapes register does not also shape factual reliability. Our estimate separates the two. A

manipulation check confirms half the responses explicitly take up the role, so the frame is received and it changes the voice; it moves accuracy about that official’s own regulation by +0.6 pp, an interval that comfortably contains zero. The frame does not change what the model knows. We tested it typed into the message rather than installed as a system instruction, which is the channel a public servant in a chat interface actually has and the narrower of the two claims (Section 5.7).

The regional model — Brazilian-Portuguese Cabra-Mistral 7B — is the *worst* performer of all fourteen ($\delta = -0.47$), so reaching for a locally tuned small model trades away the scale and frontier training that carry domain accuracy.

Practitioners deploy all three as fixes; none works, and the one a manager is most likely to try first does active damage. For Global South environmental agencies, no cheap prompt- or model-level workaround substitutes for verifying the binding value against the official source.

5.2. *The recall floor sits on the values that bind*

Accuracy collapses on the tasks that require returning a specific value in force: 0.417 for normative and local-datum recall against 0.623 for synthesis and recommendation ($\delta = -0.47$), and the floor is deepest on the national standard itself (T1, 0.367). The risk is concentrated. A manager drafting prose about air quality policy is comparatively well served; a manager retrieving the regulatory cutoff that a decision turns on is in the worst cell of the design, and worse still if they ask in a lower-resource language.

An internal check supports that this floor is genuine, not an artefact of scoring. Correcting the ground truth raised T2 — the task that had been

scored against a placeholder — from 0.368 to 0.467, while leaving T1 — which always had official ground truth — at exactly 0.367. The correction moved the task whose key was defective and left untouched the task whose key was sound.

5.3. A tier difference we can show, and a gradient we cannot

H1 was pre-specified in two parts, and the corrected scoring separates them: the tier gap holds and the monotonic gradient does not.

The gap is +5.1 pp (bootstrap CI [+1.6, +8.5]), stable to dropping any country and to every composite weighting, and a distribution-free permutation test with the country as the unit rejects the null ($p = 0.020$). Read on the composite it looks modest. Estimated task by task on the binary outcome of returning the register value, it is not: the gap scales with how much the answer depends on a fact specific to that country — odds ratio 0.22 ([0.18, 0.27]) on the binding national standard, 0.33 on national policy instruments, 0.47 on the local measured datum, 0.69 on an international health estimate, and no gap at all on open recommendation, the one task with no register value to get right. The disadvantage is concentrated, and averaging five tasks into one composite dilutes it with tasks where it does not exist. On the question a manager actually brings to these systems — what is the standard in force here — a Global South country is served at roughly a quarter of the odds of a Global North one. That is the faithful statement of the geographic finding.

The gradient fails on every reading: $\rho = 0.37$ with HDI at $n = 25$ ($p = 0.073$), Mann-Kendall agreeing ($p = 0.072$), no leave-one-out estimate approaching the pre-specified 0.55 threshold, and $\rho = 0.09$ at the registered

$n = 15$. The null is informative rather than empty, because the design has 77% power at $\rho = 0.55$: a gradient of the magnitude we pre-committed to is excluded, while a weak one is not.

The country ordering shows why the two results differ. India tops all 25 countries despite being Global South; South Africa and Indonesia outrank most of the Global North; Canada and France sit low. The disadvantage is a tier difference layered over substantial country-specific idiosyncrasy, and that idiosyncrasy is what a rank correlation over 25 countries lacks the resolution to see through.

5.4. *The mechanism, where the design has resolution*

Our pre-specified mechanism test fails, and the reason turned out to be a defect in the instrument rather than an absence of signal. The pre-specified proxy assigns the size of the Wikipedia edition in a country’s dominant official language; nine of our 25 countries — Australia, Canada, India, Kenya, Nigeria, the Philippines, the United Kingdom, the United States and South Africa — therefore receive the value of the *English* Wikipedia. The variable does not measure how much text exists in a country’s language. It measures whether the country has English as an official language, which is a fact about colonial history. We report this because it bears on any study that has used or will use the same proxy.

Between countries, neither corpus channel can be established. Country coverage correlates with accuracy at $\rho = 0.36$ ($p = 0.075$) and attenuates further after adjusting for development ($p = 0.41$), because with 25 countries coverage and development are collinear and the partial correlation exhausts the available information.

The resolution comes from changing the unit of variation rather than adding countries. Each country answers tasks that differ in how much the answer depends on a fact about *that* country, and a country’s development is constant across its own tasks. Estimating the interaction between coverage and task dependence at the response level, with a fixed effect for country, gives $\beta = +0.026$ ($p = 2 \times 10^{-8}$): greater coverage predicts a smaller specificity deficit. The association strengthens as the contrast sharpens (purest contrast $\beta = +0.042$), inverts exactly under a placebo relabelling, and survives adjustment for the language channel.

The decisive check restricts to the nine countries that share English as their official language. There the linguistic channel is not merely controlled but identical by construction, and country coverage still predicts the deficit ($\beta = +0.025$, $p = 6 \times 10^{-4}$), at the same magnitude as in the full sample. What that coefficient measures cannot be language.

We state the resulting claim at the strength the design supports and no further. Country representation is the operative channel for a gap measured in English, identified in a contrast that removes country-level confounding by construction. This is not causal identification: the design eliminates confounding at the level of the country, not at the level of task-by-country, and if the country-dependent tasks were harder in a way that happens to track coverage for an unrelated reason, the test would not distinguish that. What we can say is that the account makes a differential prediction, the prediction holds, it strengthens as the contrast sharpens, and it survives the one condition in which language is held exactly fixed.

The corpus account also appears in H2, where the unit is the language

rather than the country: the per-language penalty falls monotonically with mC4 token volume (Xue et al., 2021) and OSCAR size (Abadji et al., 2022), though with three languages this remains descriptive.

5.5. *The methodological lesson: the key matters more than the sample*

The same 9,251 responses support different substantive conclusions depending on the instrument used to score them, and the size of that difference is the lesson.

Scoring two of the five tasks against placeholder ground truth, and delegating to a language model the arithmetic of deciding whether a returned number fell inside an authorised range, yields a development gradient of $\rho = 0.51$ ($p = 0.004$) and a native-language penalty of -2.1 pp. Rebuilding those keys from official registers (WHO Ambient Air Quality Database v6.1, WHO GHO AIR_41, UNEP Global Assessment of Air Pollution Legislation) and moving the comparison into code turns the gradient into a null ($\rho = 0.37$, $p = 0.073$), more than doubles the language penalty to -4.8 pp, and converts Spanish from a null into a significant harm. The dataset did not change; only the measuring instrument did.

The field’s instinct when a geographic-bias effect proves fragile is to add countries. The ground truth deserves the scrutiny first. A defective key does not merely add noise, it adds *structured* noise, because the defects concentrate wherever official data are hardest to obtain — which is exactly where the hypothesis lives.

5.6. *Why the domain matters*

The stakes are concrete. Fine particulate matter is among the leading environmental risk factors for premature mortality, and the burden falls disproportionately on the Global South. The World Health Organization attributes 88,548 deaths in Brazil in 2021 to ambient air pollution (uncertainty interval 69,477 to 108,431), led by ischaemic heart disease, stroke and acute lower respiratory infections ([World Health Organization, 2026](#)); short-term exposure continues to drive measurable mortality in Brazilian cities ([Requia et al., 2024](#)). When a manager delegates a normative lookup or an episode recommendation to an LLM, an inaccurate answer can propagate into a regulatory brief or an emergency response.

We do not claim the model is least reliable where consequences are greatest, because we did not measure it: disease burden is not among our country covariates. The one country-level observation we have points the other way, since India carries one of the largest PM_{2.5} mortality burdens in the world and is also the highest-scoring country in our sample. Whether reliability is misaligned with need is a well-posed question, answerable by adding an age-standardised attributable-mortality covariate, but not one this design answers. We interpret the pattern we *did* measure through the coloniality-of-knowledge framework ([Mohamed et al., 2020](#)): LLMs are compressed summaries of corpora produced under uneven digital and publication access, and the language penalty is a measurable link from corpus asymmetry to knowledge asymmetry.

5.7. Limitations and future opportunities

Each limit below bounds this design, and each names the study that would remove it. We list them in the order we would run them.

Role framing was tested in the user turn, not the system prompt. The manager frame is prefixed to the message rather than installed as a system instruction, which is where the vendor guides put it. The choice follows the population we care about — public servants in the chat interface, who have no system prompt to configure — but our null bounds that user’s channel, not a developer’s. Whether a system-instruction persona recovers factual accuracy where a typed one does not is a clean two-cell experiment on the same items, and it is the first study we would run.

From panel-scored to gold-anchored. No human expert re-scored responses here. Three properties of the design bound what that costs: where the answer is a register value the verdict is computed by code and no judge is involved; where judgement is required the score is the mean of three vendors rather than one ($\text{ICC}(2, 1) = 0.56$ to $\text{ICC}(2, 3) = 0.79$); and every reported effect is a between-group contrast, in which a judge’s systematic severity cancels. Restricting to objective factual accuracy *strengthens* both headline effects, the opposite of what a permissive judge would produce, and the language effect holds on an outcome no judge touches. A blinded human-expert validation of a stratified sample would nonetheless convert an official-source-anchored result into a gold-standard-anchored one; a power analysis puts this at roughly 150 responses scored by three raters, and the released pipeline exports the sample directly.

From a within-country signal to a causal design. The channel is identified

where the design has resolution, but not causally: within countries we remove country-level confounding without removing task-by-country confounding. A design that varies country coverage while holding development fixed, or a longitudinal study tracking accuracy as a country’s encyclopedic footprint grows, would close that gap.

From a null gradient to an adequately powered test. The tier difference is established while the gradient is not, at $n = 25$. A pre-registered replication at substantially larger n — which the open dataset makes cheap — would settle whether the gradient is genuinely absent or still underpowered.

From three native languages to the full multilingual matrix. The strongest result rests on three native languages, so its *mechanism* is reported descriptively even though the effect is not. The translation-quality objection is settled empirically rather than acknowledged: a back-translation census of all 90 native prompts, audited by two models against the English originals, found none that changed the country and none that changed the requested quantity, and the penalty is unchanged among verified-faithful translations (-4.5 pp) with fidelity failing to predict its size (permutation $p = 0.42$). What remains open is breadth, not validity, and scaling to more languages is the highest-value extension this study points to.

From one domain to a cross-domain program. All effects come from one high-stakes applied domain and one corpus family (Wikimedia), and we confine the mechanistic language accordingly. The same task suite and ground-truth-registry method transfer directly to other regulated, source-anchored domains — water quality, food safety, vaccine policy.

Served-model stack as the unit of analysis. We evaluate models *as served*

through fixed access stacks, which is what a public manager queries; a residual gap could in principle reflect the stack rather than the weights. A controlled quasi-isolation design with identical open weights across stacks would separate the two, which the per-call provenance we release already enables.

5.8. *What this study contributes*

First, it is — to our knowledge — the first benchmark of LLM geographic bias in a *regulated, high-stakes applied domain* with verifiable official ground truth, moving the question beyond numeric trivia, subjective ratings, and everyday cultural knowledge to the recall and synthesis tasks a public environmental manager actually delegates. Second, it delivers a decisive and counterintuitive applied result: the three fixes practitioners reach for first — a local-manager persona, the local language, and a regionally tuned model — *all fail together*, the local language actively harms by 4.8 pp, and the regional model is the single worst performer of fourteen. This benchmark’s most useful contribution is telling the field what does not work, with effect sizes to back it. Third, it demonstrates that in this literature the ground-truth key, not the sample size, is the binding constraint: the same responses yield a significant development gradient under placeholder keys and a null one under official keys, while the language penalty more than doubles. Fourth, it separates what a 25-country design can demonstrate (a tier difference, a language penalty, a recall floor, and a corpus channel identified within countries) from what it cannot (the shape of the gradient across development, and causal identification of that channel), and reports the second set as open rather than as supported. Fifth, every reported number is recomputable from the raw data through a one-command pipeline with a built-in method-audit gate,

setting a reproducibility bar — official-source ground truth, code adjudication where the answer is a number, and end-to-end auditability — that the geographic-bias literature has largely lacked.

5.9. Implications

For environmental agencies in the Global South the operational guidance is concrete: treat LLM outputs on binding standards and measured data as drafts to be checked against the official gazette; do not assume a local-language query improves accuracy, because on this evidence it degrades it; do not expect a regionally tuned small model or a role frame to close the gap. For model developers, the language penalty is the tractable lever, and it is largest exactly where the language is least represented. For the measurement community, a geographic-bias result can turn on the quality of the ground-truth key, so the key deserves auditing before the sample is enlarged.

6. Conclusion

Large language models are becoming working infrastructure for air pollution policy, a domain where an inaccurate answer about a binding standard, a measured concentration, or an operational response carries direct public-health stakes in the Global South, where the $\text{PM}_{2.5}$ mortality burden is heaviest ([GBD 2019 Risk Factors Collaborators, 2020](#); [Requia et al., 2024](#)). We measured geographic and persona-modulated bias in that domain with a 25-country, 14-model, four-language benchmark, scored against ground truth anchored to official primary sources for every country — by code wherever the answer is a register value, and by the mean of a three-vendor judge panel elsewhere.

The remedies an agency can try do not work, and the most intuitive one backfires. Asking in the country’s own language *lowers* accuracy by 4.8 percentage points (Wilcoxon $p = 3 \times 10^{-15}$, $n = 839$ matched cells), significantly in all three languages tested and most sharply where the language is least represented in training corpora (Hindi -11.1 pp). A local-manager persona does not narrow the gap ($p = 0.27$), and the Brazilian-Portuguese regional model is the weakest of all fourteen systems ($\delta = -0.47$). Accuracy also collapses on the recall of a binding value: 0.37 on the national standard against 0.62 for synthesis and recommendation, the task a manager can least afford to have wrong.

A Global North/South tier gap persists (+5.1 pp, bootstrap 95% CI [+1.6, +8.5]), confirmed by a distribution-free permutation test with the country as the unit ($p = 0.020$). Measured task by task it is highly structured: on returning the binding national standard the Global South odds of a correct answer are 0.22 of the Global North’s ([0.18, 0.27]), and the gap shrinks monotonically as the answer depends less on a country-specific fact, disappearing on the one task with no register value to get right. It is a gap along the North/South line as conventionally drawn — partitions of the same countries on HDI alone do not reproduce it — and the monotonic development gradient does not reach significance ($\rho = 0.37$ with HDI, $p = 0.073$). We therefore report a difference between groups that we can demonstrate and a shape across the development range that we cannot.

On mechanism, the answer depends on where one looks. Between countries neither corpus channel is identifiable, because coverage and development are collinear across 25 observations. Within countries it is: comparing tasks

whose answer depends on a national fact against tasks whose answer does not — a contrast in which development is constant by construction — country coverage predicts the deficit ($\beta = +0.026$, $p = 2 \times 10^{-8}$), and it still does among the nine countries that share English as an official language, where the linguistic channel is identical by definition ($\beta = +0.025$, $p = 6 \times 10^{-4}$). We identify country representation as the operative channel without claiming causal identification.

For Global South environmental agencies the practical message is cautionary. LLM outputs on binding standards and measured data should be treated as drafts checked against the official gazette, and none of the cheap prompt- or model-level workarounds substitutes for that verification — least of all switching to the local language, which on this evidence makes the answer worse. We interpret the pattern through the colonality-of-knowledge framework (Mohamed et al., 2020): the languages under-served in training corpora (Joshi et al., 2020) are spoken where air pollution kills the most people, and this study turns that link from a plausible worry into a measured effect.

One methodological result generalises beyond this domain. Rebuilding two task keys from official registers and moving range comparison out of the judge and into code changed the substantive conclusions on the same responses, cutting the development gradient from $\rho = 0.51$ to $\rho = 0.37$ and more than doubling the native-language penalty. When an effect in this literature proves fragile, the ground-truth key deserves the audit before the sample does, because a defective key adds *structured* noise that concentrates exactly where official data are hardest to obtain — which is where the hy-

pothesis lives.

Protocol, prompts, official-source ground-truth registries, analysis code, and anonymised response data are released openly, with a one-command reproduction pipeline and a method-audit gate that recomputes every reported number from the raw data. The study also provides a reusable template for applied LLM-equity audits — official-source ground truth, code adjudication wherever the answer is a checkable number, and end-to-end reproducibility — in the high-stakes public-sector domains where these systems are now being deployed.

Data and code availability

The full protocol, prompts, analysis code, ground-truth registries, and the anonymised LLM responses will be released on publication at github.com/Roverlucas/artigo-vies-analise-regional under MIT (code) and CC-BY-4.0 (data, prompts, documentation) licenses. The release includes the per-response records that every reported effect is computed from, so each number in this paper can be recomputed from the raw data rather than taken on trust. The analysis pipeline runs end-to-end via `python code/run_all.py`.

Study protocol

Hypotheses, sampling frame, outcome definition, decision thresholds and multiplicity correction were fixed in a version-controlled analysis plan before confirmatory collection began. That plan was not deposited in a public registry, so readers cannot verify its timing independently of the repository

history, and we do not claim pre-registration. Delivered scope also departed substantially from the planned design, which the plan itself made a condition for confirmatory status.

We therefore report the study in two layers. The first states what was specified in advance and what became of it: the three co-primary hypotheses and the secondary hypothesis are reported against their declared thresholds, including the three that failed and the one whose registered direction the data reverse. The second reports what the data suggested and the plan did not anticipate, labelled exploratory throughout, with intervals and without confirmatory language. Deviations from the plan are tabulated in the supplement.

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Author contributions

Following CRediT taxonomy:

- Lucas Rover (UTFPR): Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Investigation, Visualization, Writing – Original Draft, Project Administration.
- Vitor de Melo Dominski (Faculdade Descomplica): Data Curation, Software, Formal Analysis.

- Prof. Anibal Tavares de Azevedo (Unicamp): Writing – Original Draft, Writing – Review & Editing.
- Dr. Eduardo Tadeu Bacalhau (UFPR): Validation, Supervision, Writing – Review & Editing.
- Dra. Yara de Souza Tadano (UTFPR): Validation, Supervision, Writing – Review & Editing.

Declaration of generative AI in the writing process

Two distinct uses must be separated here, because only one of them is what this declaration is about.

As an object and instrument of the study. Large language models are the subject of this research, and they also serve as one of its measuring instruments: GPT-5-mini, Gemini 2.5 Pro, Claude Sonnet 4.6 and DeepSeek-V3 scored model responses against the ground-truth registry, and Claude Sonnet 4.6 produced the native-language renderings of the prompts. These uses are part of the method, are described in Section 3, and are why the study also reports inter-judge reliability and a back-translation audit of the translated prompts. Wherever the correct answer is a value in an official register, the verdict is computed by code rather than by a model.

In preparing the manuscript. During the preparation of this work the authors used large language models to assist with drafting and editing the text and with implementing analysis code. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No language model is listed as an author, and no

model contributed the interpretation of findings or the decisions about what the evidence supports.

Declaration of interests

The authors declare no competing interests.

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